LAB REPORT: RELATIONAL JOINS & AGGREGATE FUNCTIONS

CSBB 204: Database Management Systems | Week 7 (07/09/2026 – 11/09/2026)

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Aim

To execute relational join operations (Cross Join, Equi-Join, Natural Join, Self Join, Outer Joins) and aggregate calculations (`MIN`, `MAX`, `COUNT`, `SUM`, `AVG`) grouped across relational database entities.

Theory

Concept / Function	Functional Description
`CROSS JOIN`	Produces the Cartesian product ($M \times N$) pairing every row of table A with table B.
`INNER / EQUI JOIN`	Matches rows between tables using an explicit equality predicate (`ON A.id = B.id`).
`SELF JOIN`	Joins a table to itself to compare records within the same entity relation.
`LEFT / RIGHT JOIN`	Preserves all records from the prioritized table, populating unmatched outer values with `NULL`.
`FULL OUTER JOIN`	Preserves all records from both tables; simulated in MariaDB using `LEFT JOIN UNION RIGHT JOIN`.
`Aggregate Functions`	Calculates single scalar summary values (`COUNT`, `SUM`, `AVG`, `MIN`, `MAX`) across grouped sets.

1. Cartesian product of all employees and all departments

```
MariaDB [company]> SELECT e.fname, e.lname, d.dname
    -> FROM employee e CROSS JOIN department d
    -> LIMIT 6;
+----------+--------+----------------+
| fname    | lname  | dname          |
+----------+--------+----------------+
| John     | Smith  | Headquarters   |
| John     | Smith  | Production     |
| John     | Smith  | Administration |
| John     | Smith  | Research       |
| Franklin | Wong   | Headquarters   |
| Franklin | Wong   | Production     |
+----------+--------+----------------+
6 rows in set (0.001 sec)
```

Total combinations produced = $10 \text{ employees} \times 4 \text{ departments} = 40 \text{ rows}$.

2. Differences between Cartesian Product and Join Operation

Basis	Cartesian Product (`CROSS JOIN`)	Join Operation (`INNER / EQUI`)
Predicate / Condition	No match condition required (`ON` clause omitted).	Requires an explicit matching condition (`ON` / `WHERE`).
Output Size	Always returns strictly $M \times N$ tuples.	Returns only tuples that satisfy the join condition ($M \times N$).
Practical Use	Generates exhaustive permutation matrices.	Retrieves logically linked relational data across tables.

3. All combinations of employees and projects (Cross Join)

```
MariaDB [company]> SELECT e.fname, p.pname
    -> FROM employee e CROSS JOIN project p
```

```
    -> LIMIT 6;
+-------+-----------------+
| fname | pname           |
+-------+-----------------+
| John  | ProductX        |
| John  | ProductY        |
| John  | ProductZ        |
| John  | ProductA        |
| John  | Computerization |
| John  | Reorganization  |
+-------+-----------------+
6 rows in set (0.001 sec)
```

4. Pairs of employees who work in the same department (Self Join)

```
MariaDB [company]> SELECT e1.fname AS emp1, e2.fname AS emp2, e1.dno
    -> FROM employee e1
    -> INNER JOIN employee e2 ON e1.dno = e2.dno AND e1.ssn < e2.ssn;
+----------+----------+-----+
| emp1     | emp2     | dno |
+----------+----------+-----+
| Franklin | John     |   5 |
| John     | Ramesh   |   5 |
| Franklin | Ramesh   |   5 |
| Alicia   | Jennifer |   4 |
| Alicia   | Joyce    |   4 |
| Jennifer | Joyce    |   4 |
| Ahmad    | Robert   |   1 |
| James    | Robert   |   1 |
| Ahmad    | James    |   1 |
+----------+----------+-----+
9 rows in set (0.001 sec)
```

The condition `e1.ssn < e2.ssn` eliminates duplicate inverted pairs (`(A, B)` and `(B, A)`) as well as self-pairings (`(A, A)`).

5. First name, last name, and department name of each employee (Equi-join)

```
MariaDB [company]> SELECT e.fname, e.lname, d.dname
    -> FROM employee e
    -> INNER JOIN department d ON e.dno = d.dnumber;
+----------+-----------+----------------+
| fname    | lname     | dname          |
+----------+-----------+----------------+
| Robert   | Scott     | Headquarters   |
| Ahmad    | Jabbar    | Headquarters   |
| James    | Borg      | Headquarters   |
| Anirudh  | Periasamy | Headquarters   |
| Alicia   | Zelaya    | Administration |
| Jennifer | Wallace   | Administration |
| Joyce    | English   | Administration |
| John     | Smith     | Research       |
| Franklin | Wong      | Research       |
| Ramesh   | Narayan   | Research       |
+----------+-----------+----------------+
10 rows in set (0.000 sec)
```

6. First name and project name for employees working on projects (Natural Join)

```
MariaDB [company]> SELECT e.fname, p.pname
    -> FROM employee e
```

```
    -> NATURAL JOIN (SELECT essn AS ssn, pno AS pnumber FROM works_on) w
    -> NATURAL JOIN project p
    -> LIMIT 6;
+----------+-----------------+
| fname    | pname           |
+----------+-----------------+
| John     | ProductX        |
| John     | ProductY        |
| Franklin | Computerization |
| Franklin | Reorganization  |
| Alicia   | Computerization |
| Alicia   | Allbenefits     |
+----------+-----------------+
6 rows in set (0.001 sec)
```

7. Differences between Equi-Join and Natural Join

Aspect	Equi-Join	Natural Join
Join Clause	Requires explicit `ON` or `WHERE` equality condition.	Implicitly matches on columns having identical names.
Duplicate Columns	Retains duplicate join columns from both relations.	Consolidates identical join attributes into a single column.
Column Naming	Can compare columns with completely different names.	Strictly requires attributes to share identical names.

8. Left Outer Join: Employees and department names (including unassigned employees)

```
MariaDB [company]> SELECT e.fname, e.lname, d.dname
    -> FROM employee e
    -> LEFT OUTER JOIN department d ON e.dno = d.dnumber;
+----------+-----------+----------------+
| fname    | lname     | dname          |
+----------+-----------+----------------+
| John     | Smith     | Research       |
| Franklin | Wong      | Research       |
| Robert   | Scott     | Headquarters   |
| Alicia   | Zelaya    | Administration |
| Jennifer | Wallace   | Administration |
| Ramesh   | Narayan   | Research       |
| Joyce    | English   | Administration |
| Ahmad    | Jabbar    | Headquarters   |
| James    | Borg      | Headquarters   |
| Anirudh  | Periasamy | Headquarters   |
+----------+-----------+----------------+
10 rows in set (0.000 sec)
```

9. Right Outer Join: Projects and employee names (including unassigned projects)

```
MariaDB [company]> SELECT p.pname, e.fname
    -> FROM employee e
    -> INNER JOIN works_on w ON e.ssn = w.essn
    -> RIGHT OUTER JOIN project p ON w.pno = p.pnumber
    -> WHERE p.pname = 'ProductA' OR p.pnumber ⩽ 3;
+----------+-------+
| pname    | fname |
+----------+-------+
| ProductX | John  |
| ProductX | Joyce |
| ProductY | John  |
| ProductY | Joyce |
```

```
| ProductZ | Ramesh|
| ProductA | NULL  |
+----------+-------+
6 rows in set (0.001 sec)
```

`ProductA` is preserved despite having no assigned workers, yielding a `NULL` employee value.

10. Full Outer Join: All employees and all projects

```
MariaDB [company]> SELECT e.fname, p.pname
    -> FROM employee e
    -> LEFT JOIN works_on w ON e.ssn = w.essn
    -> LEFT JOIN project p ON w.pno = p.pnumber
    -> UNION
    -> SELECT e.fname, p.pname
    -> FROM project p
    -> LEFT JOIN works_on w ON p.pnumber = w.pno
    -> LEFT JOIN employee e ON w.essn = e.ssn
    -> LIMIT 7;
+----------+-----------------+
| fname    | pname           |
+----------+-----------------+
| John     | ProductX        |
| John     | ProductY        |
| Robert   | NULL            |
| Anirudh  | NULL            |
| NULL     | ProductA        |
| Franklin | Computerization |
| Franklin | Reorganization  |
+----------+-----------------+
7 rows in set (0.001 sec)
```

MariaDB emulates `FULL OUTER JOIN` using `UNION` between corresponding `LEFT` and `RIGHT` queries.

11. SSNs of employees who work on project number 1 (Inner Join)

```
MariaDB [company]> SELECT DISTINCT e.ssn, e.fname, e.lname
    -> FROM employee e
    -> INNER JOIN works_on w ON e.ssn = w.essn
    -> WHERE w.pno = 1;
+-----------+-------+---------+
| ssn       | fname | lname   |
+-----------+-------+---------+
| 123456789 | John  | Smith   |
| 453453453 | Joyce | English |
+-----------+-------+---------+
2 rows in set (0.000 sec)
```

12. Left Outer Join: Departments and projects starting with 'P'

```
MariaDB [company]> SELECT d.dname, p.pname
    -> FROM department d
    -> LEFT OUTER JOIN project p ON d.dnumber = p.dnum AND p.pname LIKE 'P%';
+----------------+----------+
| dname          | pname    |
+----------------+----------+
| Headquarters   | NULL     |
| Production     | ProductA |
| Administration | NULL     |
| Research       | ProductX |
| Research       | ProductY |
| Research       | ProductZ |
```

```
+---------------+----------+
6 rows in set (0.000 sec)
```

Placing `AND p.pname LIKE 'P%'` inside the `ON` clause preserves departments having no matching projects as `NULL` instead of eliminating them.

13. Multiple joins: Employee, department, and project names (including unassigned employees)

```
MariaDB [company]> SELECT e.fname, e.lname, d.dname, p.pname
    -> FROM employee e
    -> LEFT JOIN department d ON e.dno = d.dnumber
    -> LEFT JOIN works_on w ON e.ssn = w.essn
    -> LEFT JOIN project p ON w.pno = p.pnumber
    -> LIMIT 7;
+----------+---------+----------------+-----------------+
| fname    | lname   | dname          | pname           |
+----------+---------+----------------+-----------------+
| John     | Smith   | Research       | ProductX        |
| John     | Smith   | Research       | ProductY        |
| Franklin | Wong    | Research       | Computerization |
| Franklin | Wong    | Research       | Reorganization  |
| Robert   | Scott   | Headquarters   | NULL            |
| Alicia   | Zelaya  | Administration | Computerization |
| Alicia   | Zelaya  | Administration | Allbenefits     |
+----------+---------+----------------+-----------------+
7 rows in set (0.001 sec)
```

14. Department name and average salary for employees in each department

```
MariaDB [company]> SELECT d.dname, ROUND(AVG(e.salary), 2) AS avg_salary
    -> FROM department d
    -> INNER JOIN employee e ON d.dnumber = e.dno
    -> GROUP BY d.dname;
+----------------+------------+
| dname          | avg_salary |
+----------------+------------+
| Administration |   32780.00 |
| Headquarters   |   53613.33 |
| Research       |   39680.00 |
+----------------+------------+
3 rows in set (0.000 sec)
```

15. Summary statistics of all employee salaries

```
MariaDB [company]> SELECT
    ->   SUM(salary)        AS total_salary,
    ->   MAX(salary)        AS max_salary,
    ->   MIN(salary)        AS min_salary,
    ->   ROUND(AVG(salary), 2) AS avg_salary
    -> FROM employee;
+--------------+------------+------------+------------+
| total_salary | max_salary | min_salary | avg_salary |
+--------------+------------+------------+------------+
|    376780.00 |   70040.00 |   24500.00 |   41864.44 |
+--------------+------------+------------+------------+
1 row in set (0.000 sec)
```

16. Count of employees working on projects in the 'Research' department

```
MariaDB [company]> SELECT COUNT(DISTINCT w.essn) AS research_proj_workers
    -> FROM works_on w
```

```
    -> INNER JOIN project p ON w.pno = p.pnumber
    -> INNER JOIN department d ON p.dnum = d.dnumber
    -> WHERE d.dname = 'Research';
+-----------------------+
| research_proj_workers |
+-----------------------+
|                     3 |
+-----------------------+
1 row in set (0.001 sec)
```

17. Number of employees in each department

```
MariaDB [company]> SELECT d.dname, COUNT(e.ssn) AS num_employees
    -> FROM department d
    -> LEFT OUTER JOIN employee e ON d.dnumber = e.dno
    -> GROUP BY d.dname;
+----------------+---------------+
| dname          | num_employees |
+----------------+---------------+
| Administration |             3 |
| Headquarters   |             4 |
| Production     |             0 |
| Research       |             3 |
+----------------+---------------+
4 rows in set (0.000 sec)
```

18. Maximum salary in each department

```
MariaDB [company]> SELECT d.dname, MAX(e.salary) AS max_salary
    -> FROM department d
    -> INNER JOIN employee e ON d.dnumber = e.dno
    -> GROUP BY d.dname;
+----------------+------------+
| dname          | max_salary |
+----------------+------------+
| Administration |   49340.00 |
| Headquarters   |   70040.00 |
| Research       |   45200.00 |
+----------------+------------+
3 rows in set (0.000 sec)
```